Marium Zeeshan

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Computer Science student focused on Generative AI, multimodal systems, and full-stack development. Experienced building scalable solutions from low-level systems to LLM-powered applications, with a strong interest in applied AI research.

Education

Habib University, Karachi *2022–2026*
BS Computer Science 100% HUTOPS Scholar
Relevant Coursework: NLP, Data Science, Computer Vision, Digital Image Processing, Deep Learning, Web Development

Aga Khan Higher Secondary School *2019 – 2021*
Pre-Engineering High Achiever: 92.5%

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, Verilog
AI / ML & GenAI: PyTorch, Transformers, RAG Pipelines, Multimodal Systems (ViLT, BLIP), LangChain
Web Development: React.js, Node.js, Express.js, Vite, Tailwind CSS, REST APIs, MongoDB, MySQL, MSSQL
Data Science: NumPy, Pandas, Matplotlib, Seaborn, SHAP, EDA
Tools & Platforms: Git/GitHub, MATLAB, Vivado, SQL Server Management Studio, WordPress, Docker (basic)

Experience

JDC Foundation *June 2026 – Sep 2026*
- Developed a React-based Mortuary Management System handling service bookings, admin approvals, and automated notifications.
- Implemented multi-tier role-based access control (Customer, Admin, Head Admin) with persistent browser-based storage.
- Engineered a bilingual UI supporting real-time English and Urdu script localization for streamlined community operations.

Optimizers Lab (Pvt.) Ltd. | Product Development Intern *Feb 2026 – May 2026*
- Designed and developed responsive websites with improved UI/UX and performance optimisation using HTML, CSS, and JavaScript.
- Implemented server-side logic, database integration, and REST API connections for web applications.
- Managed WordPress themes, plugins, and CI/CD deployment workflows for multiple client projects.

10Pearls | Data Science Intern *Summer 2025*
- Developed a Weather and AQI Prediction System using the Open-Meteo API and Scikit-learn regression models for environmental forecasting.
- Built dynamic dashboards for real-time data visualisation, improving stakeholder accessibility to environmental metrics.

AlBarkah Bank | IT Intern *Summer 2025*
- Engineered a SQL-based database-driven alerting system to monitor service health and system threshold breaches.
- Analysed mobile banking architecture and authentication flows to support secure digital banking operations.

Publications

HU at SemEval-2026 Task 10: Psycholinguistic Conspiracy Marker Extraction and Detection *Jul 2026*
Co-author | SemEval 2026 (ACL Anthology)
- Ranked 1st globally in the SemEval-2026 Task 10 evaluation.
- Implemented Global Pointer architecture with T5, DeBERTa, and GLiNER2 for span-based NER, extracting 5 fine-grained psycholinguistic markers.
- Boosted conspiracy text classification to 0.80 F1 score across imbalanced benchmarks by fine-tuning T5 with custom focal loss and benchmarking against CT-BERT-PRCT and DistilBERT baselines.

AI & Research Projects

Multilingual Multimodal Agricultural Chatbot | Final Year Project (FYP) *Ongoing*
Industry Collaboration with Pakistan Agriculture Research (PAR)
- Designed a scalable multilingual RAG-based LLM pipeline over 4TB of multimodal data (text, images, audio, video) integrating Qdrant vector search and GPT-4 mini API for real-time agricultural knowledge retrieval.
- Processed multimodal inputs using Whisper Large with VAD for speech transcription, Google Cloud Vision API for OCR, and OpenAI API for English-Urdu translation.
- Optimised context-aware response generation and inference efficiency through advanced prompt engineering and retrieval augmentation techniques.

Interpretable Anomaly Detection via VLM Fine-Tuning *Jan 2026 – May 2026*
- Benchmarked LLaVA, LLaMA 3.2-Vision, and Qwen2.5-VL; fine-tuned 4-bit LoRA on 9,005-image MVTec AD/LOCO benchmark to eliminate output format errors.
- Applied GRPO reinforcement alignment with a custom reward function, reaching 0.93 macro-F1 and 4.3/5.0 LLM-as-a-judge score (Top 3 cohort).

Visual Question Answering (VQA) System *2025*
- Built a multimodal pipeline using ViLT and BLIP architectures to generate natural language answers from visual inputs.

Technical Projects

Completely Fair Scheduler (CFS) – OS Kernel Modification
- Modified the xv6 kernel to implement a Completely Fair Scheduler (CFS) ensuring equitable CPU time distribution across processes.
- Developed a custom `setnice()` system call for dynamic process priority adjustment.

Cinema Booking System using Red-Black Trees
- Implemented a cinema seat booking engine using red-black trees for O(log n) data retrieval and seating management.

Blood Donor App
- Built a full-stack MERN application for blood donor registration and real-time donation request fulfilment.

CortechTalent Corporate Website
- Designed and deployed a responsive, mobile-first corporate website using HTML, CSS, and JavaScript with secure form handling.

LUDO Game – OOP Implementation
- Developed a fully functional LUDO game in C++ using inheritance, polymorphism, and abstraction with modular class design.